

Appendix I: Survey documentation

Table A1: Misperception questionnaire

Dimension	Question	Variable Name	Rationale
Population (I)	What do you think: What is the percentage of people in the Swiss population that were born abroad? (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Immigrants Estimate: NICS8a_wp2 Confidence: NICS8b_wp2	Standard misperception item commonly used in previous surveys. Note that we refined the question wording to avoid the 1-in-X-bias.
Population (II)	What do you think: What is the percentage of refugees among the people who immigrated to Switzerland last year? By refugees we mean people who have applied for asylum in Switzerland and subsequently received a residence permit on humanitarian grounds. (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Refugees Estimate: NICS9a_wp2 Confidence: NICS9b_wp2	More specific innumeracy item focused on the most discursively prominent immigrant sub-group.
Culture (I)	What do you think: What is the percentage of Muslims among foreigners living in Switzerland? (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Muslims Estimate: NICS10a_wp2 Confidence: NICS10b_wp2	Religious affiliation is a standard measure of culture—here it might be even more relevant, as cultural conceptions of being Muslim and a range of related stereotypes are rather widespread. Moreover, the literature suggests that immigrants are often perceived as Muslims and vice versa (Lutz and Bitschnau 2023).
Culture (II)	What do you think: What is the percentage of people from the European Union among foreigners living in Switzerland? (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Europeans Estimate: NICS11a_wp2 Confidence: NICS11b_wp2	Country or region of origin often functions as a cultural marker. In the Swiss context, immigrants are usually categorised as either from the European Union or from so-called <i>third countries</i>. EU nationals constitute the majority of the Swiss immigrant population and benefit from privileged access to the Swiss labour market.
Economy (I)	What do you think: What is the unemployment rate of foreigners living in Switzerland? (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Unemployed Estimate: NICS12a_wp2 Confidence: NICS12b_wp2	Standard item reflecting the common assumption that immigrants are seen as a burden on the welfare state.
Economy	What do you think: What is the	Educated	Education-related item that contrasts

(II)	percentage of people with a university degree among foreigners living in Switzerland? (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Estimate: NICS13a_wp2 Confidence: NICS13b_wp2	with the unemployment item. It is grounded in Switzerland's role as a popular destination for high-skilled individuals and their families.
Security (I)	What do you think: What is the percentage of foreigners among all people who were convicted last year in Switzerland? (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Criminals Estimate: NICS14a_wp2 Confidence: NICS14b_wp2	Item addressing perceived security threats, using the percentage of immigrants among those sentenced as a proxy for the share of crimes committed by them.
Security (II)	What do you think: What is the percentage of undocumented immigrants among all foreigners living in Switzerland? (Please enter a number between 0 and 100.) How sure are you of your answer? (Response scale from 0 - Not at all sure to 10 - Completely sure)	Undocumented Estimate: NICS15a_wp2 Confidence: NICS15b_wp2	Item addressing the governability of immigration, using the share of undocumented immigrants as a proxy for the state's capacity to control immigration.

Note: This table provides the English translations of the survey questions fielded in German, French and Italian.

Source list of benchmark statistics

(1) Immigrants

Statistical concept: Share of foreign-born individuals in the permanent resident population.

Official value: 31% (2022).

Note: The figure has been rounded up from 30.6%. The percentage of foreign-born individuals is slightly higher than the percentage of non-citizens, which is another common statistical reference. However, the share of the foreign-born is more directly related to immigration and also allows for better international comparability.

Source: Federal Statistical Office (2024) Permanent and non-permanent resident population by canton, residence permit, place of birth, sex and age, 2010–2023 [FSO number px-x-0103010000_102].

(2) Refugees

Statistical concept: Share of refugees in all new immigrant arrivals into the permanent resident population.

Official value: 5% (2022).

Note: We combine three statistical categories (recognised refugee status after asylum granted, hardship cases as part of the asylum process, FNA regulation cases related to asylum) to measure the number of

refugees. This total is then divided by the number of all new arrivals in the foreign permanent resident population: $(3,677 + 4,332 + 243) / 162,435 = 5.08\%$ (rounded down to 5%)

Source: State Secretariat for Migration (2023) Foreign Population and Asylum Statistics 2022. Bern: Swiss Government.

(3) Muslims

Statistical concept: Share of Muslims in the foreign permanent resident population.

Official value: 14% (2022).

Note: This figure is based on public statistics on religious affiliation by nationality within the permanent resident population. It gives the proportion of Muslims among all foreign permanent residents: $250,726 / 1,789,817 = 14.01\%$, with a confidence interval of ± 0.8 percentage points.

Source: Federal Statistical Office (2023) Religionszugehörigkeit nach Nationalität [FSO number su-d-01.08.02.03].

(4) Europeans

Statistical concept: Share of European Union citizens in the foreign permanent resident population.

Official value: 66% (2022).

Note: This figure is based on annual migration statistics and represents the proportion of citizens of European Union member states among all foreign permanent residents: $1,484,005 / 2,241,854 = 66.20\%$.

Source: State Secretariat for Migration (2023) Foreign Population and Asylum Statistics 2022. Bern: Swiss Government.

(5) Unemployed

Statistical concept: Official unemployment rate among foreign permanent residents.

Official value: 4% (2022).

Note: This figure has been rounded up from 3.8%. It is based on the official Swiss unemployment rate, which differs from the ILO unemployment rate but is the primary statistic reported on this subject.

Source: State Secretary for Economic Affairs (2023) Arbeitslosigkeit in der Schweiz 2022. Bern: Federal Statistical Office.

(6) Educated

Statistical concept: Share of university graduates among the foreign permanent resident population.

Official value: 35% (2022).

Note: This figure has been rounded up from 34.7% and is based on the working age population (ages 25 to 64): 34.7%. The total share of tertiary graduates is higher, as it also includes individuals with higher professional education.

Source: Federal Statistical Office (2024) Bildungsstand der Bevölkerung – Daten des Indikators [FSO number ind-d-406101].

(7) Criminals

Statistical concept: Share of criminals without Swiss citizenship among all convicted criminals.

Official value: 54% (2022).

Note: This figure is based on statistics distinguishing between convicted minors and adults. We combine these categories to calculate the overall percentage of non-Swiss convicted criminals: $61,004 / 112,294 = 54.33\%$. This includes not only foreigners in the permanent resident population but also individuals still applying for asylum or not residing in Switzerland (e.g., tourists). The figure does not adjust for factors such as age, which makes the item more cognitively demanding and subject to varied interpretations.

Sources: Federal Statistical Office (2024) Jugendurteile und verurteilte Personen, Schweiz und Kantone [FSO number je-d-19.03.02.03.02.01.02a]; Erwachsene: Verurteilungen und Verurteilte für ein Vergehen oder Verbrechen, Schweiz und Kantone [FSO number je-d-19.03.02.02.02.01.02a].

(8) Undocumented

Statistical concept: Share of foreigners without a valid residence permit among all foreigners living in Switzerland.

Estimated value: 4% (2015).

Note: This figure represents the proportion of *sans-papiers* living in Switzerland in 2015 (76,000) relative to the total population of *sans-papiers* and legal foreign residents (i.e., $76,000 + 1,993,916 = 2,069,916$, with $76,000 / 2,069,916 = 3.67\%$). Two important caveats apply: (1) the data is significantly older than the rest, and (2) because undocumented immigrants are nowhere registered, it is itself based on an estimation with a likely range of 58,000 to 105,000 *sans-papiers* (Morlok et al. 2015). An earlier estimate from 2005 is similar, suggesting that the number of undocumented immigrants has remained rather stable over time. It seems therefore reasonable to assume that the true value is indeed close to 4%.

Source: Michael Morlok, Harald Meier, Andrea Oswald, Denise Efionayi-Mäder, Didier Ruedin, Dina Bader, Philippe Wanner (2015) Sans-Papiers in der Schweiz 2015. Studie von B,S,S. Volkswirtschaftliche Beratung AG im Auftrag des Staatssekretariats für Migration.

Report on quantitative cognitive tests (pretest phase)

Given the novelty of the chosen approach and the complexity of the task, the authors and the MOSAiCH team conducted a series of tests to assess the validity of the items and refine the questionnaire design accordingly. Our aim was not only to examine respondents' understanding of the cognitive dimension (i.e., the heuristics used to estimate percentages) but also to identify the optimal wording and response format (i.e., whether the variance is sufficiently large when the real value is small). To achieve this, we implemented quantitative cognitive tests through an online survey. In addition, after half of the items, we included an open-ended question asking respondents to describe how they arrived at their answers, followed by a three-point scale question assessing whether the option to provide a certainty value had made the estimation task easier, more difficult, or neither. The question module was fielded in an online panel, with around 1440 respondents.

Overall, these tests proved quite valuable in identifying how respondents provide both estimates and certainty values. The results indicate that the questions were generally understood as intended and thus yield valid measurements for analysis. The testing also drew attention to a question where the wording

caused ambiguity. Specifically, we examined (a) an alternative phrasing of the main respondent task, (b) the clarity of individual items, and (c) whether the certainty measure was perceived as helpful.

As regards the alternative phrasing of the main task, our rationale was informed by the European Social Survey (ESS 2016), which had fielded a similar item. The wording of the ESS (“Of every 100 people, how many would you say are...”) does not use the term percentage because it was thought to be a cognitive burden—but we found that respondents were frequently confused by this formulation (“Do you mean percentage?” “What exactly is the difference to percentage?”). When we compared this version to a less complicated alternative (“What percentage of all persons living in Switzerland...”), the latter was clearly better understood. For this reason, we opted to include it in the survey.

Regarding the clarity of the items, we focused in particular on the feasibility of asking respondents to provide estimates. In particular, we examined whether respondents would attempt to look up the real values before answering. Our quantitative results suggest that most respondents did not engage in this time-consuming task: response variance was high, and most estimates deviated substantially from the actual values. Our qualitative results corroborated these findings. Only a few respondents admitted that they had looked up specific figures; most reported relying on everyday experiences and perceptions to inform their estimates.

The tests were also used to assess the clarity of our questions, i.e., whether their wording is easily and correctly understood. Here, both qualitative and quantitative results indicated that the vast majority of respondents interpreted the questions as intended. However, we identified an issue with the phrasing of the question on the percentage of foreigners among those convicted of a crime (NICS14a_wp2), which was not statistically precise and came along with considerable ambiguity in both the statistical reference and the corresponding formulation. As a consequence, we revised the question from “Wie viele Prozent der in der Schweiz lebenden Personen, die im vergangenen Jahr für eine Straftat verurteilt worden sind, sind Ausländerinnen oder Ausländer?” to “Wie viel Prozent der Personen, die im vergangenen Jahr in der Schweiz wegen einer Straftat verurteilt wurden, sind Ausländerinnen oder Ausländer?”

Finally, we evaluated whether respondents understood the task of providing a certainty value for each item and whether they followed our instructions. The qualitative data indicated that the task was clearly understood, and the quantitative data further supported this conclusion: On the one hand, respondents used the scale to indicate varying levels of certainty depending on the item. On the other hand, only 7% of the respondents reported that providing certainty values made their task harder, suggesting that this additional question was not perceived as overly burdensome or distracting.

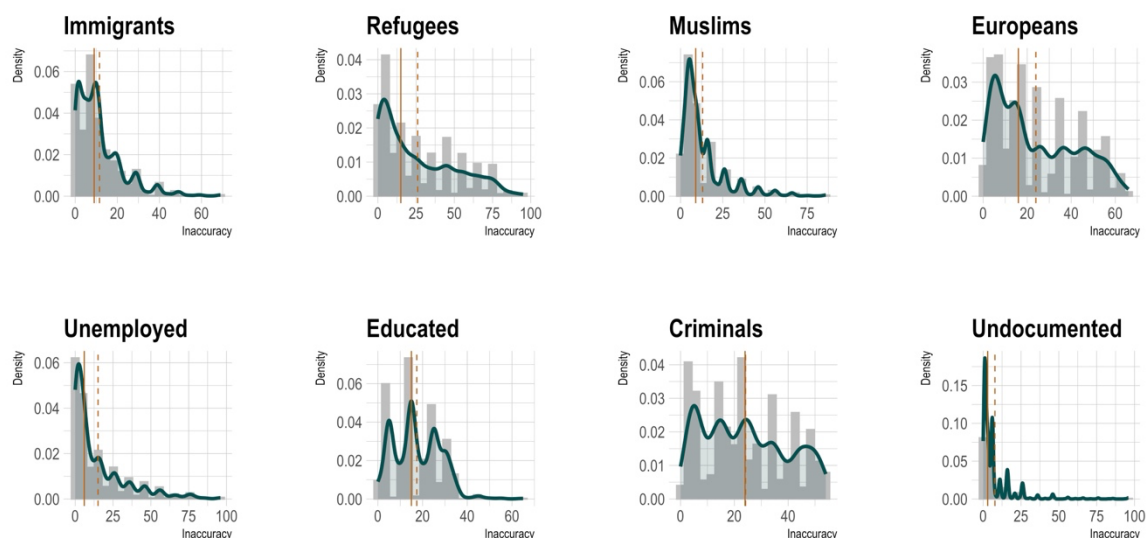
All these items have been integrated into the follow-up questionnaire (Part 2), which was administered both online and on paper, and comprised 16 questions for respondents.

Source: Stähli et al. (2024: 31–32)

Appendix II: Additional analyses

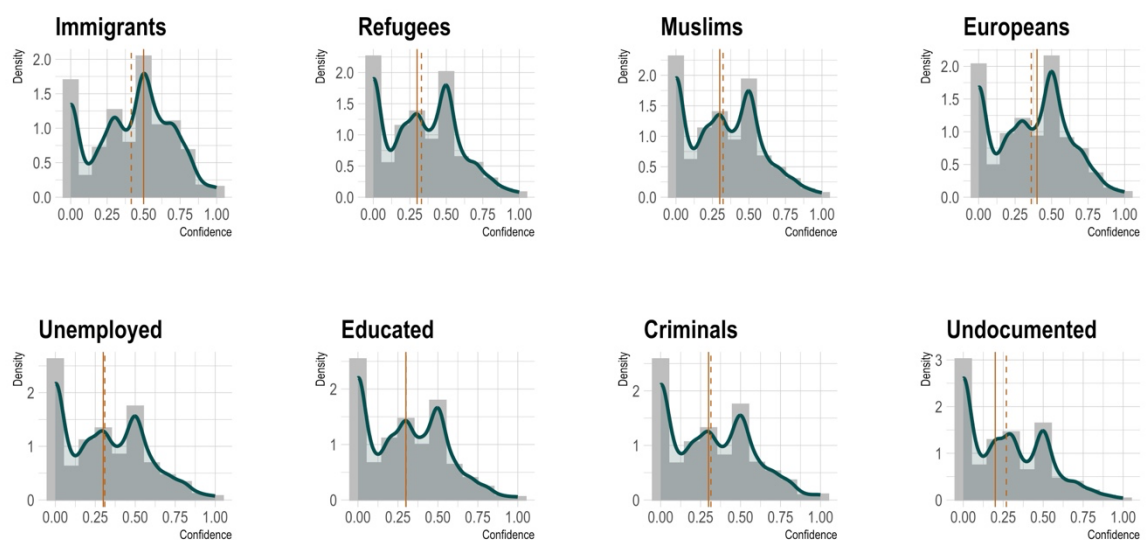
Descriptive statistics

Figure A1: Distribution of inaccuracy of immigration perceptions



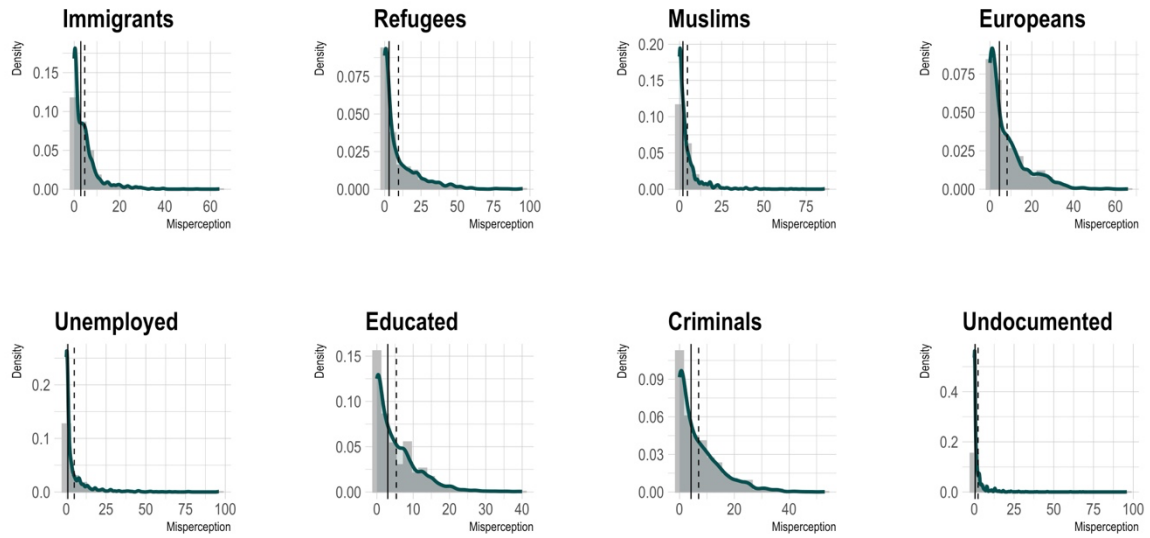
Note: Histograms combined with density plots of perception inaccuracy (absolute distance from benchmark value). Summary statistics are presented in black (dashed line: mean, solid line: median).

Figure A2: Distribution of confidence in immigration perceptions



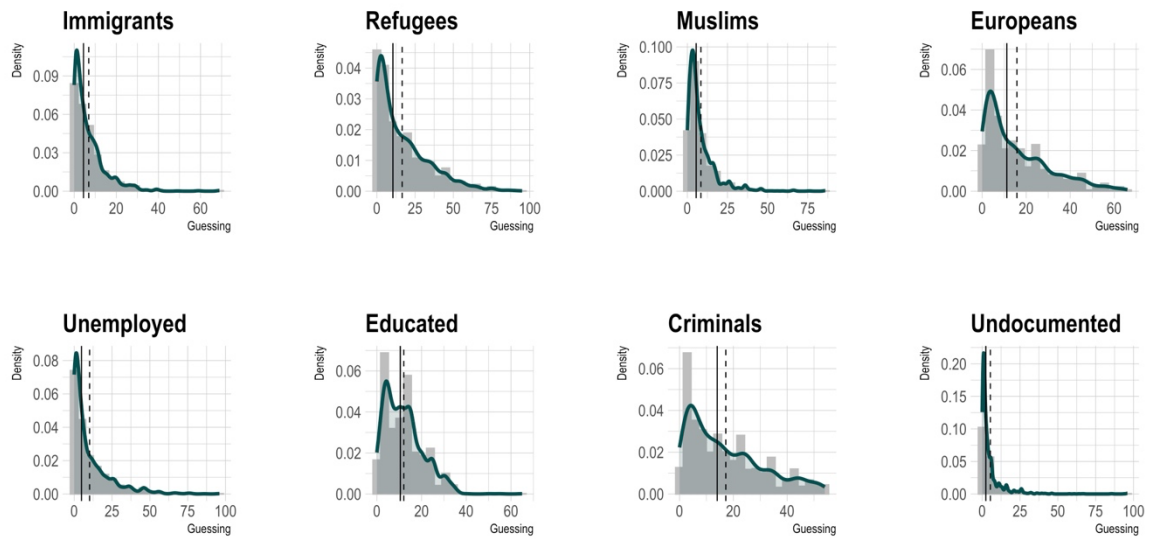
Note: Histograms combined with density plots of perception confidence. Summary statistics are presented in black (dashed line: mean, solid line: median).

Figure A3: Distribution of misperception scores



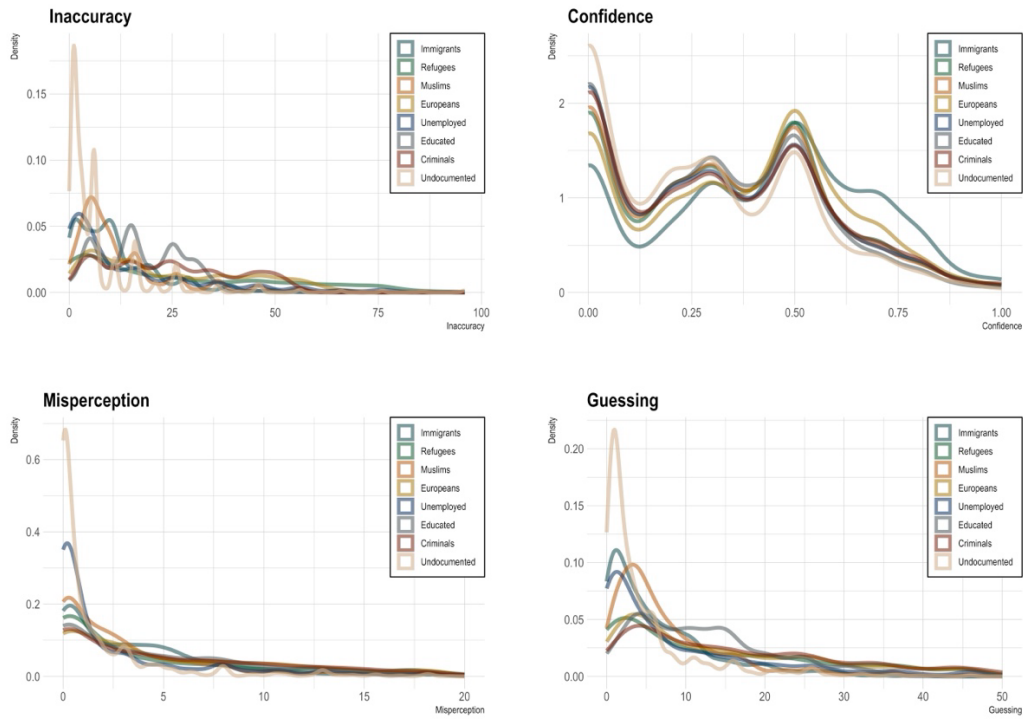
Note: Histograms combined with density plots of misperception scores. These scores are calculated as the product of the inaccuracy of a perception and the confidence in it. Summary statistics are presented in black (dashed line: mean, solid line: median).

Figure A4: Distribution of guessing scores



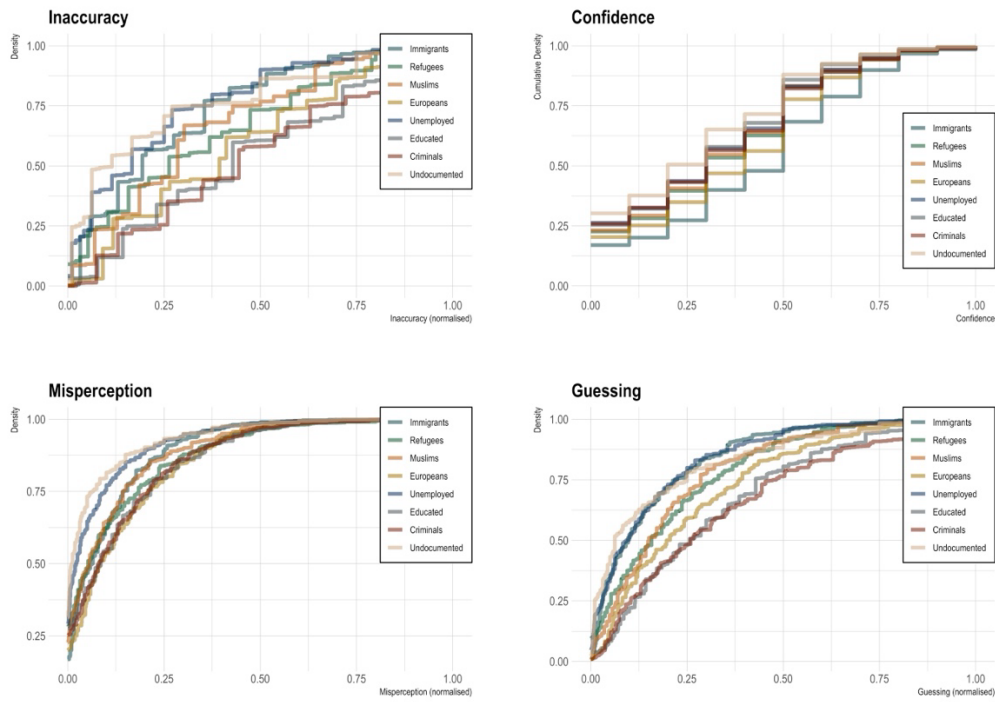
Note: Histograms combined with density plots of guessing scores. These scores are calculated as the product of the inaccuracy of a perception and the lack of confidence in it. Summary statistics are presented in black (dashed line: mean, solid line: median).

Figure A5: Density plots



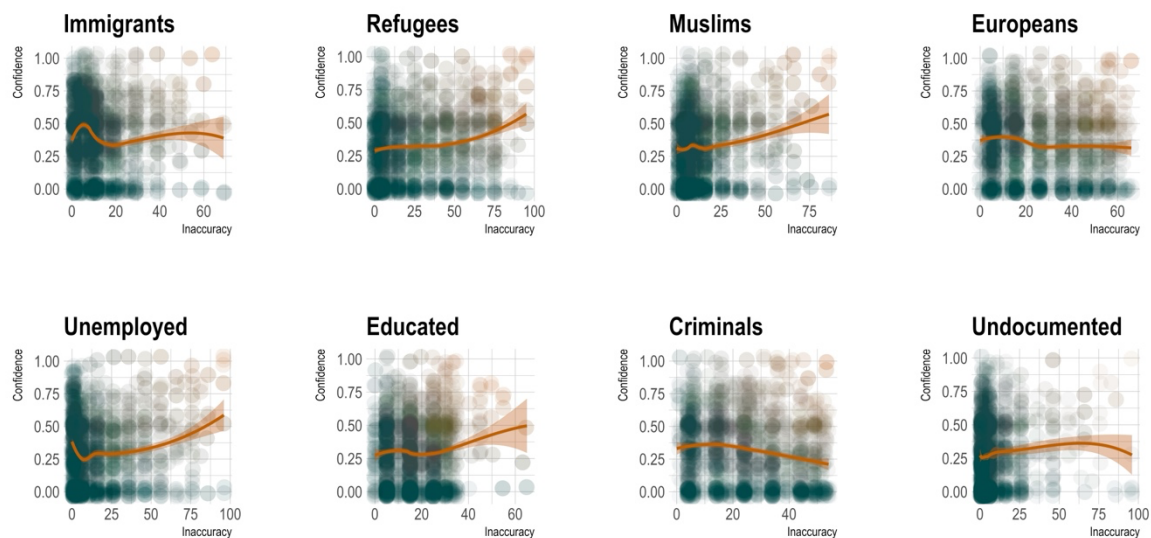
Note: Density plots equivalent to Figure 1 of the manuscript, using regular density rather than cumulative density.

Figure A6: Cumulative density plots with normalized scores



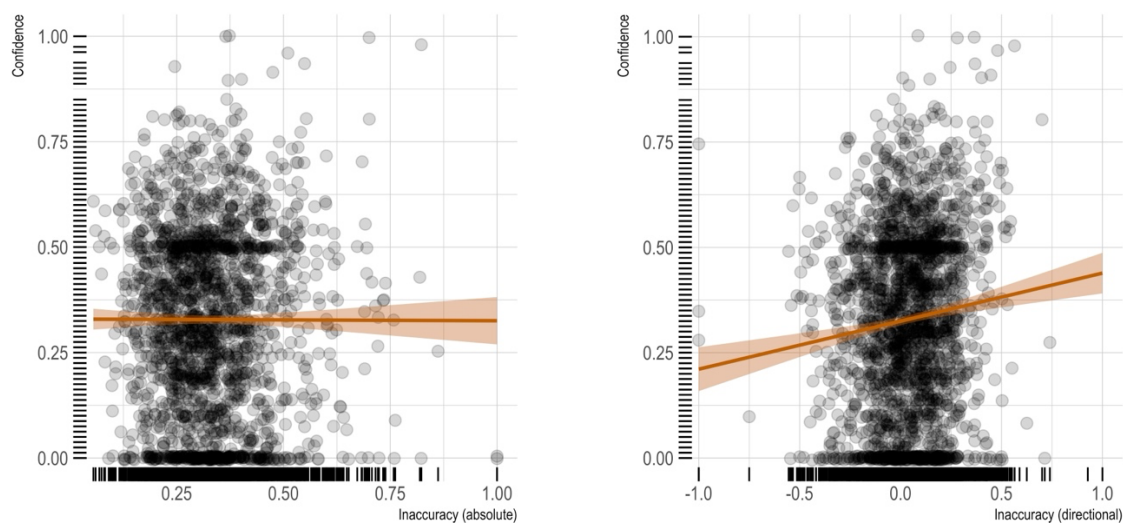
Note: Cumulative density plots of immigration perceptions. These plots are equivalent to Figure 1 in the manuscript, but use normalized scores for inaccuracy, misperception, and guessing—that is, values are adjusted relative to the maximum possible error. The x-axes for Misperception and Guessing are restricted to the area containing substantive data to enhance visibility.

Figure A7: Scatterplots of inaccuracy and confidence



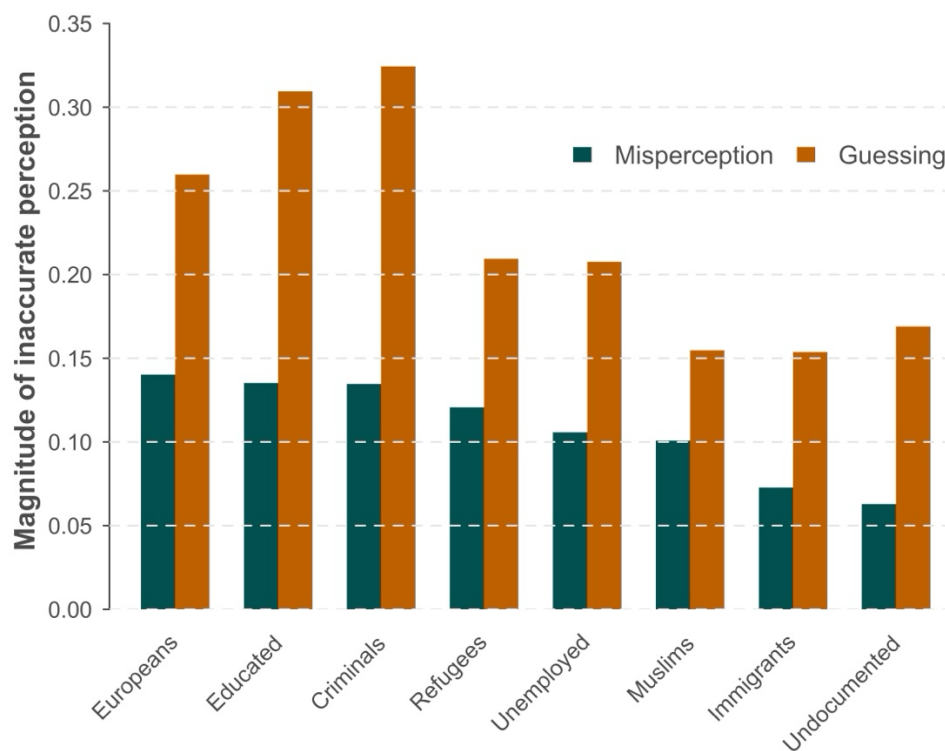
Note: Multiple scatterplots showing the inaccuracy of and confidence in perceptions. Values are jittered and semi-transparent to reduce overplotting. The color gradient indicates the misperception score (more orange corresponds to greater misperception). A LOESS curve with a 95% confidence interval displays the bivariate association.

Figure A8: Scatterplots of inaccuracy and confidence (aggregated)



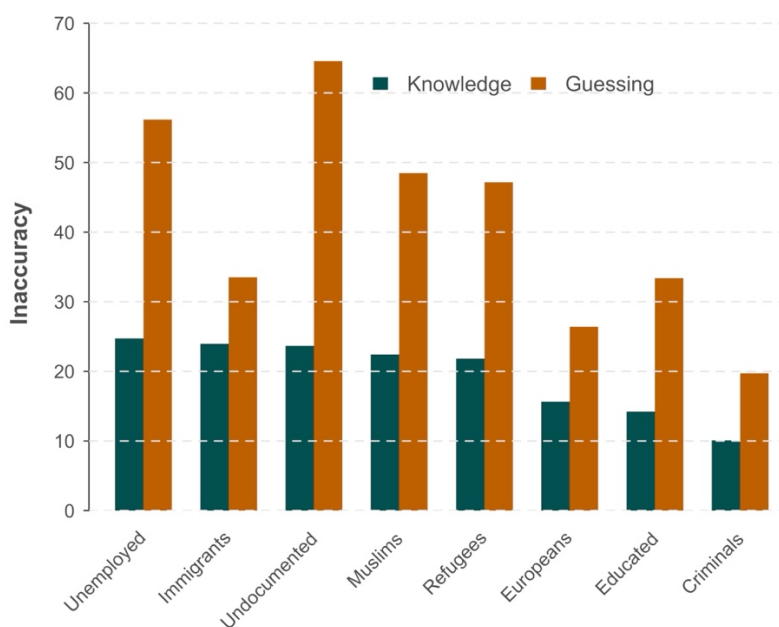
Note: Scatterplots showing the relationship between confidence and inaccuracy based on mean values across perception items. The left plot uses normalized absolute deviation from the benchmark value, whereas the right plot uses normalized directional deviation from it. The overall pattern remains consistent even when outliers are excluded.

Figure A9: Average misperception and guessing based on normalized scores



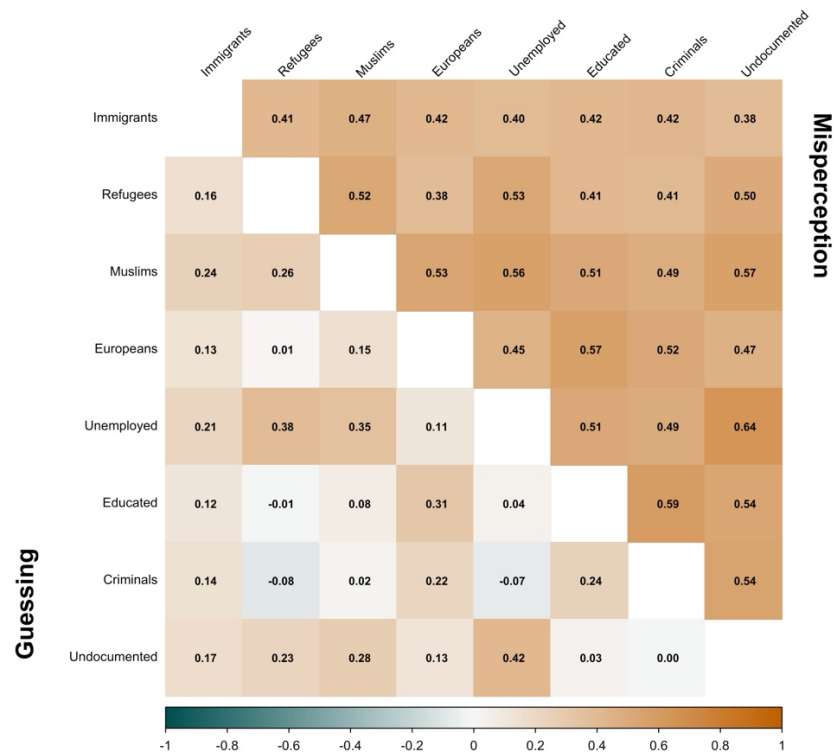
Note: Bar plots showing average misperception and bad guessing scores, sorted in descending order by misperception.

Figure A10: Average knowledge and (good) guessing



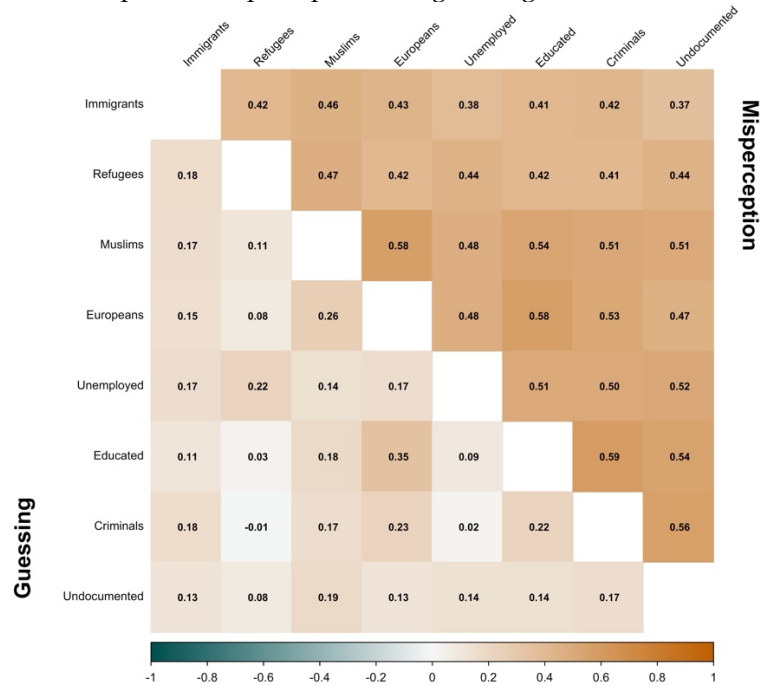
Note: Bar plots showing average knowledge and good guessing scores, sorted in descending order by knowledge.

Figure A11: Correlation plot of misperception and guessing scores



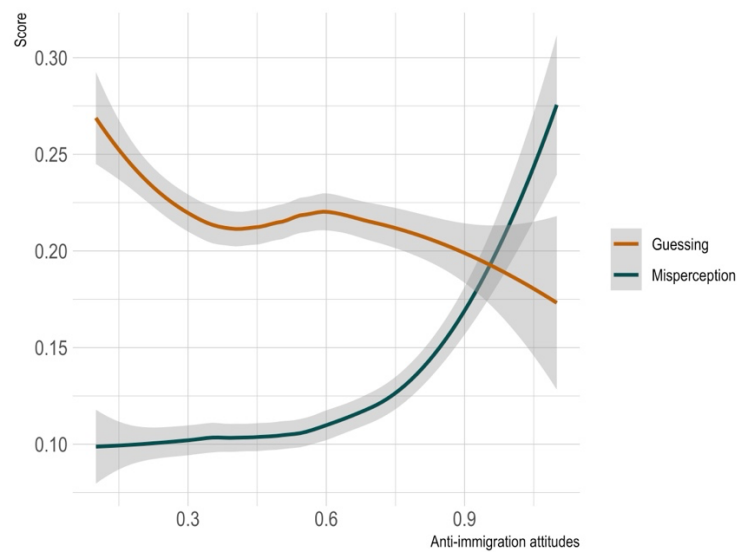
Note: Visual representation of a correlation matrix using Spearman correlation coefficients, chosen due to the highly skewed distribution of the scores.

Figure A12: Correlation plot of misperception and guessing scores based on normalized scores



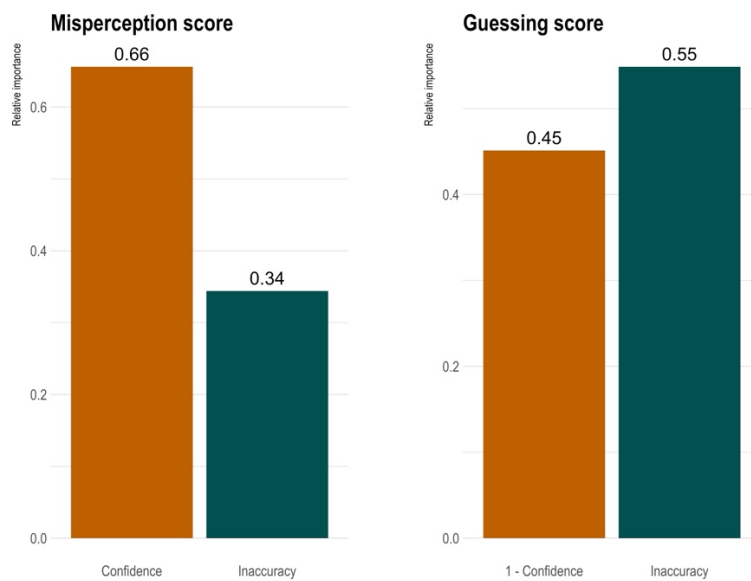
Note: Visual representation (normalized) of a correlation matrix using Spearman correlation coefficients, chosen due to the highly skewed distribution of the scores.

Figure A13: Normalized misperception and guessing by immigration attitudes



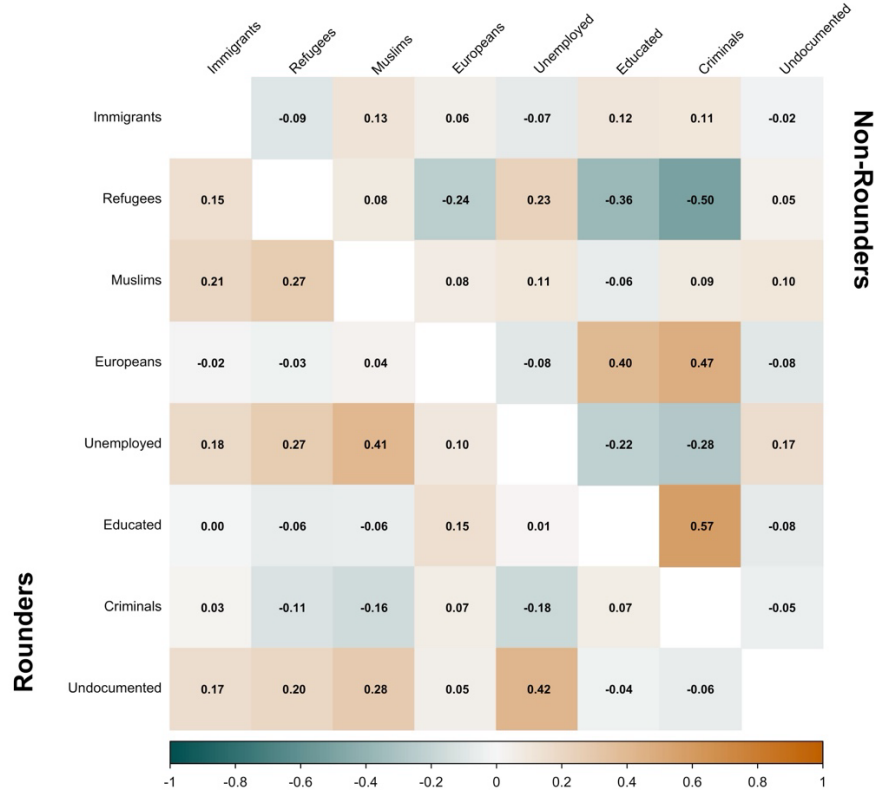
Note: Bivariate association between misperception, guessing, and anti-immigration attitudes, shown with LOESS curves. Shaded areas represent 95% confidence intervals.

Figure A14: Variance decomposition of misperception and guessing scores



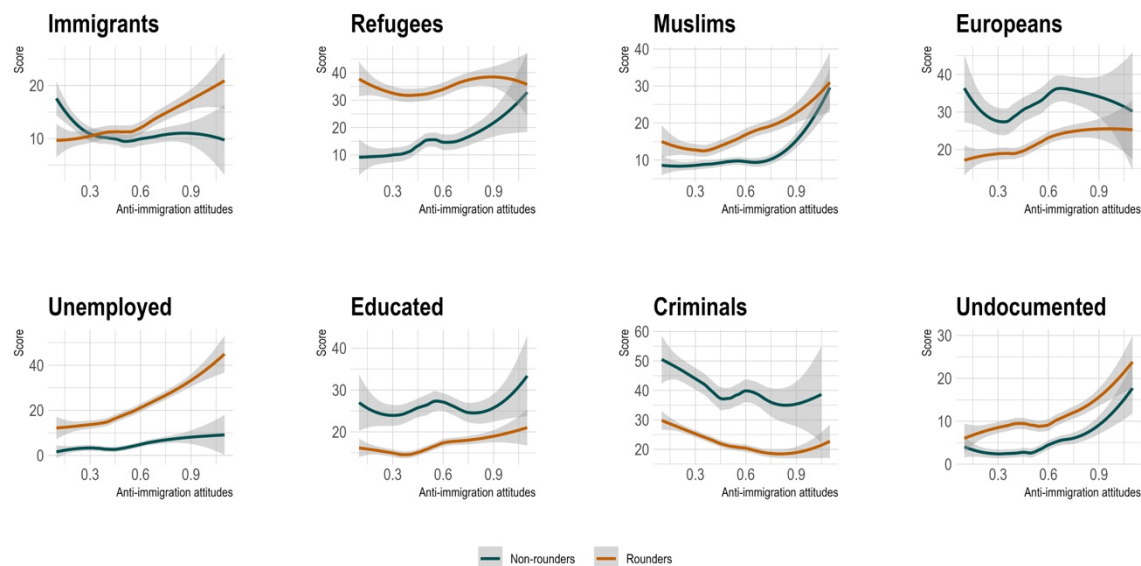
Note: Bar plots showing the relative importance of the two defining components, measured using the "lmg" metric from Shapley Value Regression.

Figure A15: Correlation plots of immigration perceptions for rounders and non-rounders



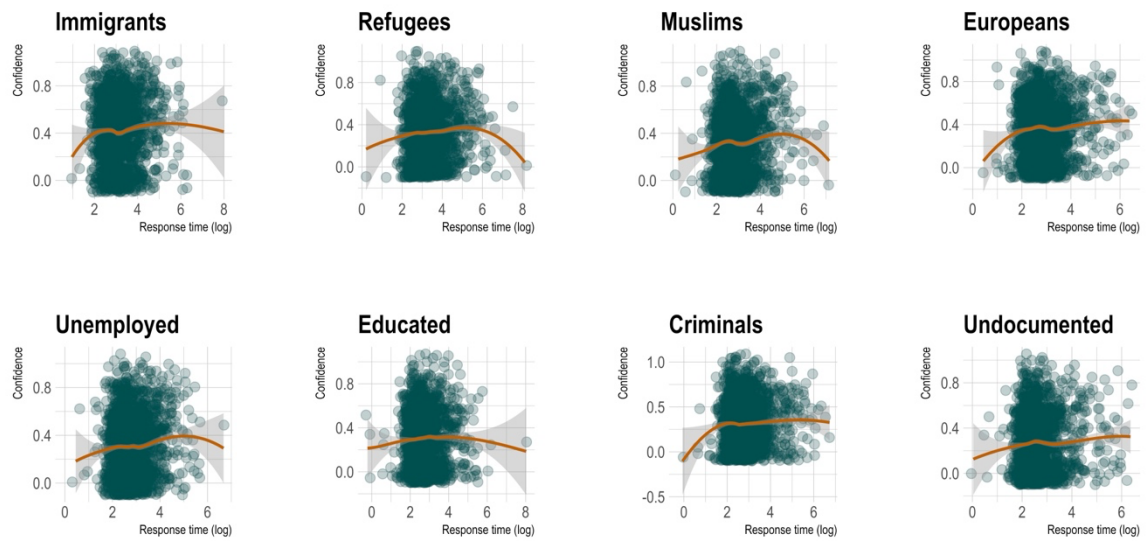
Note: Visual representation of a correlation matrix using Spearman correlation coefficients, chosen due to the skewed distribution of the variables.

Figure A16: Association between rounding and immigration attitudes



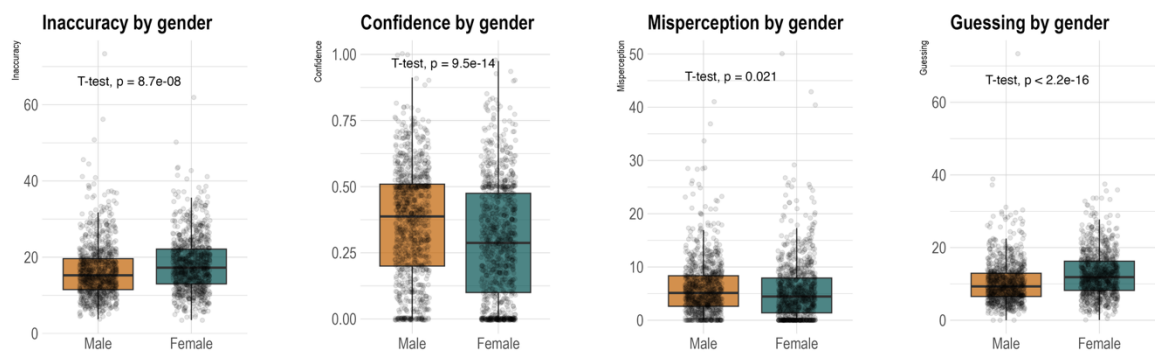
Note: Bivariate association between the inaccuracy of (non-)rounders and respondents' immigration attitude, displayed with (LOESS) curves. Shaded areas represent 95% confidence intervals.

Figure A17: Scatterplots between confidence and response time



Note: Bivariate association between respondents' confidence and the log-transformed time taken to complete the estimation task, displayed with LOESS curves. Shaded areas represent 95% confidence intervals.

Figure A18: Box plots for gender differences



Note: Boxplots comparing the main variables of interest by gender (male/female), including mean-difference t-tests.

Table A2: Factor loadings and communalities for latent trait models of accuracy and confidence

Perception item	Accuracy loading	Accuracy h ²	Confidence loading	Confidence h ²
Immigrants	0.935	0.874	0.854	0.729
Refugees	0.888	0.788	0.939	0.883
Muslims	0.947	0.897	0.963	0.928
Europeans	0.969	0.939	0.955	0.911
Unemployed	0.923	0.852	0.950	0.902
Educated	0.944	0.890	0.941	0.886
Criminals	0.858	0.737	0.880	0.775
Undocumented	0.727	0.529	0.933	0.870

Note. Loadings from one-factor models using standardized input variables. SS Loadings: Accuracy = 6.506; Confidence = 6.884. Proportion of Variance Explained: Accuracy = 81.3%; Confidence = 86.1%.

Interpretation of the composite scores

Example for misperception:

- Actual Value: 50%
- Estimate: 60%
- Inaccuracy: $|60\% - 50\%| = 10$
- Confidence: 0.8
- Misperception Score: $10 \times 0.8 = 8.0$

Example for guessing:

- Actual Value: 50%
- Estimate: 60%
- Inaccuracy: $|60\% - 50\%| = 10$
- Confidence: 0.8
- Misperception Score: $10 \times (1 - 0.8) = 2.0$